

MIZAN TEPI UNIVERSITY
COLLEGE OF COPUTING AND INFORMATION(CCL)
DEPARTMENT OF INFORMATION TECHNOLOGY
COURSE TITLE NETWORKING PROJECT

PREPARED BY GROUP 3 SECTION A

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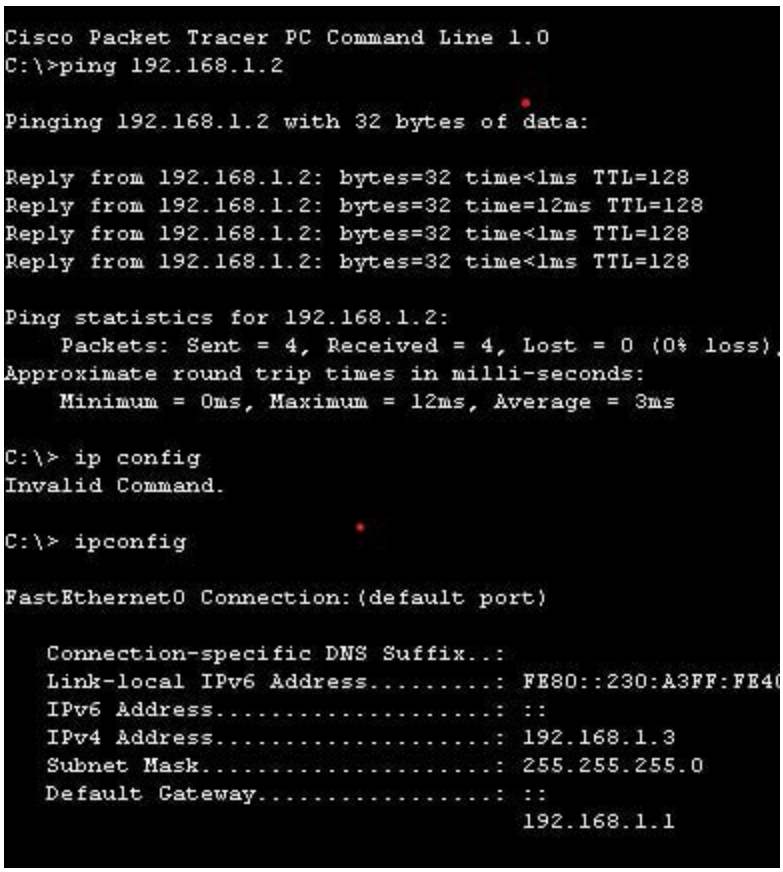
SUBMITED TO:Mr Merahgn

Introduction

As organizations grow, the need to separate network resources for different departments becomes crucial. A flat network where all devices can communicate freely creates significant security risks and network performance bottlenecks due to excessive broadcast traffic. The provided network diagram illustrates a common scenario where two departments share the same physical infrastructure but require logical separation and controlled connectivity..

Problem Statement

The current network infrastructure faces challenges related to:
Security: Lack of logical separation allows any user to access resources in any department, risking data breaches.
Performance: All devices operate in a single broadcast domain, leading to high network congestion and potential broadcast storms.
Cost: While multiple physical networks would solve separation issues, they are costly to implement and manage. A cost-effective solution using existing hardware is required.
1 TEST CONictivity
PC0-PC1



PC2- PC3

```
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time=3ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time=3ms TTL=128

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 1ms

C:\> ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: FE80::290:CFF:FEB3:B924
   IPv6 Address.....: ::
   IPv4 Address.....: 192.168.2.2
   Subnet Mask.....: 255.255.255.0
   Default Gateway.....: ::
                                 192.168.2.1
```

PC1 TO PC3

```
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: FE80::2E0:A3FF:FE37:5E94
   IPv6 Address.....: ::
   IPv4 Address.....: 192.168.1.2
   Subnet Mask.....: 255.255.255.0
   Default Gateway.....: ::
                                 192.168.1.1
```

2. Give reason why its not communicating even if they are on same class?

Reasons for Lack of Communication (Same Class)
Even if computers are in the same IP Class (e.g., Class C), they will not communicate if:
Different SUBNET/Networks: Department IT might be 192.168.1.2/24 and Dept CS might be 192.168.2.3/24. Different SUBNT WORK require a router to communicate.
Missing Default Gateway: The computers do not know where to send traffic destined for a different network.
Firewall Settings: Windows Firewall or security software might be blocking ICMP (ping) traffic.

3. What will be the possible way to create communication between each Department?

To enable communication between two different departmental networks, you must use a Router or a Layer 3 Switch.
1.Connect both department switches to a router.
2.Assign an IP address to each router interface to act as the Default
3.Gateway for each department.Configure PCs to use the router interface as their default gateway.

4. Create your own network and make this two departments to communicate each other.

4. Create Own Network Design (Example)
Here is a setup to make Department IT and Department CS communicate using Cisco Packet Tracer
Network Layout:
Dept IT : PC0 (192.168.1.2), PC1 (192.168.1.3) connected to Switch 1.
Dept CS: PC2 (192.168.2.2), PC3 (192.168.2.3) connected to Switch 1.
Router: Connect Switch 1 to Gig0/0 (IP: 192.168.1.1) and Switch 1 to Gig0/1 (IP: 192.168.2.1).
Configuration:
PC0 & PC1 Gateway: 192.168.1.1
PC2 & PC3 Gateway: 192.168.2.1
With this, PC1 can ping PC3 via the router.

```
Switch>enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name IT
Switch(config-vlan)#EXIT
Switch(config)#vlan 20
Switch(config-vlan)#name CS
Switch(config-vlan)#EXIT
Switch(config)#interface range fa0/2-11
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport  access vlan 10
Switch(config-if-range)#no shutdown
Switch(config-if-range)#exit
Switch(config)#interface range fa0/12-23
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport  access vlan 20
Switch(config-if-range)#no shutdown
Switch(config-if-range)#exit
Switch(config)# exit
```

```
Switch#show vlan brief

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Fa0/1, Fa0/24, Gig0/1, Gig0/2
10   IT                                active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                 Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                 Fa0/10, Fa0/11
20   CS                                active    Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                 Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                 Fa0/20, Fa0/21, Fa0/22, Fa0/23
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active
Switch#enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#interface fa0/24
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
```

```
Switch#
Switch#enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/24
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 10,20
Switch(config-if)#exit
```

```
Router>
Router>enable
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

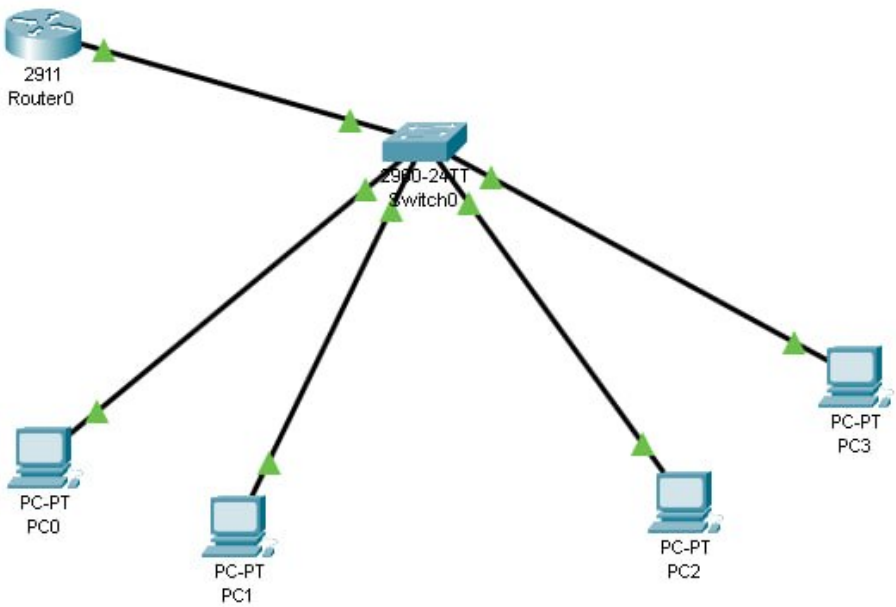
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gig0/0.10
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

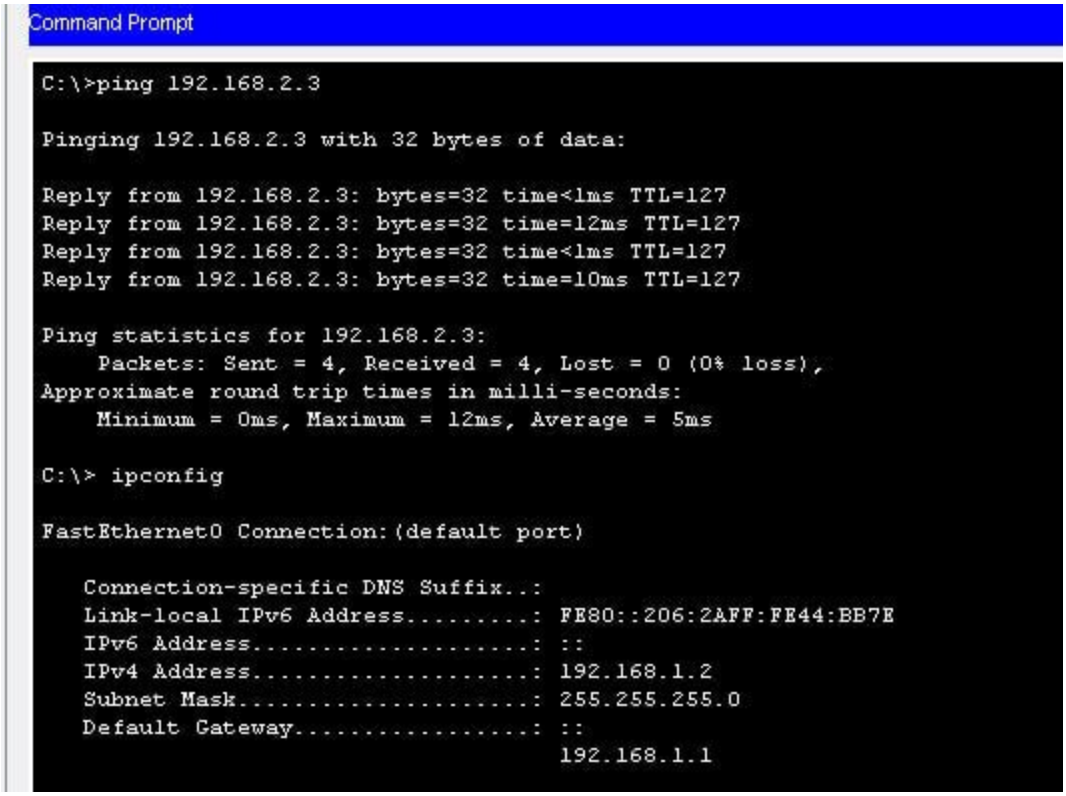
Router(config-subif)#interface gig0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.1.1  255.255.255.0
Router(config-subif)#exit
Router(config)#interface gig0/0.20
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.20, changed state to up

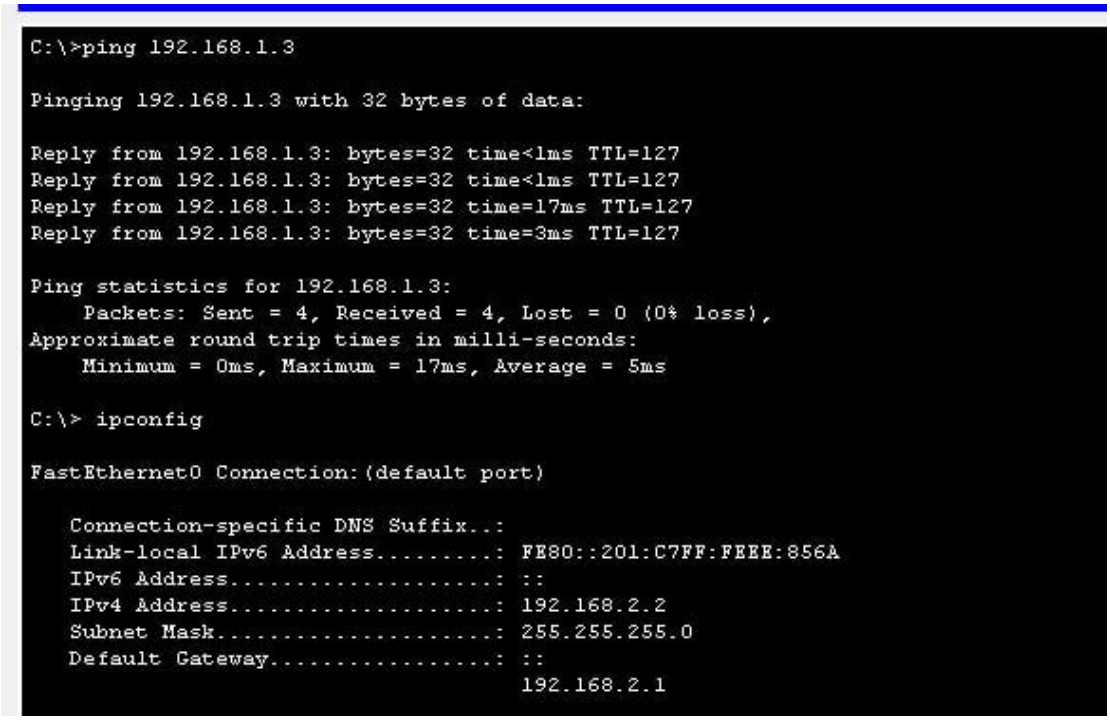
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config-subif)#interface gig0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.2.1  255.255.255.0
Router(config-subif)#exit
Router(config)#
```



PC0-PC2 REPLAY


Command Prompt
C:\>ping 192.168.2.3
Pinging 192.168.2.3 with 32 bytes of data:
Reply from 192.168.2.3: bytes=32 time<1ms TTL=127
Reply from 192.168.2.3: bytes=32 time=12ms TTL=127
Reply from 192.168.2.3: bytes=32 time<1ms TTL=127
Reply from 192.168.2.3: bytes=32 time=10ms TTL=127
Ping statistics for 192.168.2.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 12ms, Average = 5ms
C:\> ipconfig
FastEthernet0 Connection:(default port)
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: FE80::206:2AFF:FE44:BB7E
IPv6 Address.....: ::
IPv4 Address.....: 192.168.1.2
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
192.168.1.1



PC1-PC3 REPLAY

5. Mention your requirements for question No 4

Requirements for Question No. 4
Hardware: 4 PCs, 1 Switches (e.g., 2960), 1 Router, Copper Straight-Through cables.
IP Addressing: Two distinct subnets
(e.g., 192.168.1.1/24 and 192.168.2.1/24).
Software: Simulation software (e.g., Packet Tracer) to configure interfaces.
Configuration: Configured Default Gateways on all PCs and routing enabled on the Router

Conclusion

Connectivity between different departments is only possible through Layer 3 routing. While a switch connects devices within the same network, a Router is required to bridge different subnets. By assigning a Default Gateway to each PC, you provide a path for data to travel across network boundaries, turning isolated departments into a fully functional, interconnected network